

9 More on TST

9.1 Eyring transition-state theory

The form of transition-state theory which we have derived above looks quite different from Eyring TST which we used in the first lecture. Unlike our new version, Eyring TST is not rigorously derived but instead obtained using thermodynamic arguments. Here we will show how it is related to the classical TST above.

First transform from the old coordinates $\vec{q} = (q_1, \dots, q_{3N})$ to new coordinates $(s, u_1, \dots, u_{3N-1})$ and from old momenta $\vec{p} = (p_1, \dots, p_{3N})$ to $(p_s, w_1, \dots, w_{3N-1})$. The coordinate transformation should be orthogonal such that $d\vec{p}d\vec{q} = dsd\vec{u}dp_sdw$. They are mass-weighted such that the associated mass for the $(\vec{u}, \vec{w})$ degrees of freedom are all the same, m .

$$k_{\text{TST}}(T) = (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \int e^{-\beta p_s^2/2m_s - \beta \vec{w}^2/2m} \frac{p_s}{m_s} \Theta(p_s) dp_s d\vec{w} \int e^{-\beta V(s, \vec{u})} \delta(s - s^\ddagger) ds d\vec{u} \quad (9.1)$$

$$= \frac{1}{2\pi\beta\hbar} \frac{Q_{s^\ddagger}(T)}{Q_{\text{r}}(T)} \quad (9.2)$$

where we have integrated over s and p_s and

$$Q_{s^\ddagger}(T) = (2\pi\hbar)^{-(3N-1)} \int e^{-\beta \vec{w}^2/2m} d\vec{w} \int e^{-\beta V(s^\ddagger, \vec{u})} d\vec{u} \quad (9.3)$$

is the classical partition function constrained to the dividing surface.

Optimizing the dividing-surface to maximize the rate forces it to pass through regions of high potential energy and the optimum dividing surface will thus often pass through (or at least close to) a saddle point on the potential-energy surface, where $\nabla V(\vec{q}) = 0$. This saddle point is often known as the transition-state and the barrier height, $V^\ddagger$ is the difference in energy between this point and the potential of the reactants in their minimum-energy configuration. In Eyring's TST, we choose the dividing surface to pass through this saddle point such that the surface is perpendicular to the unstable normal mode corresponding to the imaginary frequency.

Shifting the energy of this constrained partition function by the barrier height, we define the transition-state partition function relative to the barrier height energy,

$$Q^\ddagger(T) = Q_{s^\ddagger}(T) e^{\beta V^\ddagger} \quad (9.4)$$

and therefore

$$k_{\text{TST}}(T) = \frac{k_{\text{B}}T}{h} \frac{Q^\ddagger(T)}{Q_{\text{r}}(T)} e^{-\beta V^\ddagger} \quad (9.5)$$

This looks like (but isn't) the traditional Eyring TST formula which you already know. The difference is that here the partition functions are evaluated by a classical phase-space average whereas Eyring's formula uses quantum partition functions within the rigid-rotor and harmonic-oscillator approximations. Note that this doesn't make Eyring's TST "quantum" because the treatment along the s coordinate is still completely classical.

The underlying assumptions of Eyring's TST are that

- the Born-Oppenheimer approximation is valid
- there is little or no recrossing of the dividing surface
- the reaction coordinate can be separated from all other coordinates
- the dynamics is classical along the reaction coordinate
- the rigid-rotor and harmonic oscillator approximations hold for the other degrees of freedom

9.2 Langevin dynamics

For a one-dimensional system (phase space p, x , mass m , potential $V(x)$) interacting with a harmonic bath, we define a system-bath model:

$$H = \frac{p^2}{2m} + \sum_j \frac{p_j^2}{2} + V(x, q)$$

with the total potential-energy surface

$$V(x, q) = V(x) + \sum_j \frac{1}{2} \omega_j^2 \left(q_j - \frac{c_j}{\omega_j^2} x \right)^2 \quad (9.6)$$

By expanding the square, it can be written in the alternative form:

$$V(x, q) = V(x) + \frac{1}{2} \lambda x^2 - \sum_j c_j q_j x + \sum_j \frac{1}{2} \omega_j^2 q_j^2 \quad \text{where } \lambda = \sum_j \frac{c_j^2}{\omega_j^2} \quad (9.7)$$

Applying Hamilton's equations of motion gives

$$\begin{aligned} \dot{x} &= \frac{p}{m} & \dot{p} &= -\nabla V(x) + \sum_j c_j \left(q_j - \frac{c_j}{\omega_j^2} x \right) \\ \dot{q}_j &= p_j & \dot{p}_j &= -\omega_j^2 q_j + c_j x. \end{aligned}$$

Unless $V(x)$ is a very simple function, these cannot be solved explicitly. However, if we imagine that $x(t)$ were known, then we can solve for the dynamics of the bath. These are just inhomogeneous differential equations which have the solution:

$$q_j(t) = q_j(0) \cos \omega_j t + p_j(0) \frac{\sin \omega_j t}{\omega_j} + c_j \int_0^t x(t') \frac{\sin \omega_j(t-t')}{\omega_j} dt'$$

Show that this is a solution and then integrate by parts to show that it can be written equivalently as

$$q_j(t) - \frac{c_j}{\omega_j^2} x(t) = \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right) \cos \omega_j t + p_j(0) \frac{\sin \omega_j t}{\omega_j} - c_j \int_0^t \frac{p(t')}{m} \frac{\cos \omega_j(t-t')}{\omega_j^2} dt'.$$

Substituting into the formulas above, we get:

$$\dot{p} = -\nabla V(x) - \int_0^t \frac{p(t')}{m} K(t-t') dt' + R(t), \quad (9.8)$$

where we have defined the friction kernel and random force:

$$K(t-t') = \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j(t-t')$$

$$R(t) = \sum_j c_j \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right) \cos \omega_j t + \sum_j c_j p_j(0) \frac{\sin \omega_j t}{\omega_j}$$

Of course the force is not really random, but if there are many bath modes and we do not know their initial conditions exactly then it appears random. You could see this for yourself by selecting $q_j(0)$ and $p_j(0)$ from a random distribution and plotting the function against t .

To analyse the friction kernel further, we go to the continuum limit (i.e. assume the bath is made up of an infinite number of modes) and replace the sum by an integral weighted by the density of states

$$K(t-t') = \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j(t-t')$$

$$\rightarrow \int_0^\infty g(\omega) \frac{c(\omega)^2}{\omega^2} \cos \omega(t-t') d\omega$$

In fact if $g(\omega) = \sum_j \delta(\omega - \omega_j)$ then these expressions are clearly identical. However, we can now also choose a smooth function for $g(\omega)$ which is easier to deal with analytically.

In fact we don't even have to specify $g(\omega)$ and $c(\omega)$ independently, but can write everything in terms of the spectral density $J(\omega) = \frac{\pi}{2} g(\omega) \frac{c(\omega)^2}{\omega}$. We will take the special case that this follows the so-called Ohmic form $J(\omega) = \gamma \omega$. Then

$$\begin{aligned} K(t-t') &= \frac{2}{\pi} \int_0^\infty \frac{J(\omega)}{\omega} \cos \omega(t-t') d\omega \\ &= \frac{2}{\pi} \int_0^\infty \gamma \cos \omega(t-t') d\omega \\ &= \frac{1}{\pi} \int_{-\infty}^\infty \gamma \cos \omega(t-t') d\omega \quad \text{because it is an even function} \\ &= \frac{1}{\pi} \int_{-\infty}^\infty \gamma e^{i\omega(t-t')} d\omega \quad \text{because the imaginary part integrates to 0} \\ &= 2\gamma \delta(t-t') \end{aligned}$$

Therefore,

$$\dot{p} = -\nabla V(x) - \gamma \frac{p}{m} + R(t)$$

which is the Langevin dynamics equation. You should recognize the term $-\gamma \frac{p}{m} = -\gamma \dot{x}$ as being the simplest type of friction force and which points opposite to the velocity. The random force describes Brownian motion, which have probably also seen before.

Using the equipartition theorem,* it is easy to show that

$$\begin{aligned} \left\langle q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right\rangle &= 0 & \langle p_j(0) \rangle &= 0 \\ \left\langle \left(q_j(0) - \frac{c_j}{\omega_j^2} x(0) \right)^2 \right\rangle &= \frac{kT}{\omega_j^2} & \langle p_j(0)^2 \rangle &= kT \end{aligned}$$

where we have assumed that the bath is in thermal equilibrium for a given value of $x(0)$. Then we can show that

$$\begin{aligned} \langle R(t)R(t') \rangle &= \sum_j \frac{kT c_j^2}{\omega_j^2} \cos \omega_j t \cos \omega_j t' + \sum_j kT c_j^2 \frac{\sin \omega_j t}{\omega_j} \frac{\sin \omega_j t'}{\omega_j} \\ &= kT \sum_j \frac{c_j^2}{\omega_j^2} \cos \omega_j(t - t') \\ &= kTK(t - t') = 2kT\gamma\delta(t - t'). \end{aligned}$$

This means that the random force at one time t is completely independent of the random force at another time $t' \neq t$. The strength of the random force is proportional to $\sqrt{2kT\gamma}$ and is thus dependent on temperature. It is in some ways similar to a standard gaussian random variable ξ_i , which has the property $\langle \xi_i \xi_j \rangle = \delta_{ij}$, which is known as white noise. However, there are also some subtleties due to the fact that $R(t)$ is not differentiable with respect to time. For a numerical implementation of the Langevin equation, see the homework exercise and Tuckerman's textbook.

9.3 Kramers theory

Transition state theory can be easily applied to a one-dimensional double well problem and gives

$$k_{\text{TST}} = (2\pi\beta\hbar)^{-1} Q_r(T)^{-1} e^{-\beta V^\ddagger}$$

where $Q_r(T)$ is the reactant partition function and $V^\ddagger$ is the barrier height. Using classical mechanics and a harmonic approximation we find $Q_r(T) = 1/(\beta\hbar\omega_0)$ where ω_0 is the frequency of the reactant well.

Kramers argued that for Langevin dynamics in a one-dimensional double well problem, recrossing would cause the classical rate with strong friction to be lower than TST and that it would decrease as friction got stronger. However, he also predicted that for very weak friction, there would also be recrossing such that the classical rate would decrease as friction got weaker. In between there must be a “turnover” regime where the rate is neither decreasing nor increasing. In the homework exercise you can simulate this behaviour by running Langevin trajectories to compute $C_{\text{fs}}(t)$.

*This says that the thermal average of the kinetic-energy or potential-energy terms of a harmonic oscillator are always equal to $kT/2$.

Here we can study the strong friction regime using a multidimensional TST approach on the full system-bath model. It is unfortunately not so easy to study the weak friction regime in this way as TST just simply cannot describe this type of recrossing.

For simplicity, we only take one bath mode, but the analysis can be extended fairly straightforwardly to a large bath. We are interested in the transition state, which we can define as the saddle point. This is seen to be located at $x = 0$ and $q_1 = 0$. A second-order Taylor expansion around this point gives the system-bath potential:

$$V(x, q_1) = V^\ddagger - \frac{1}{2}\bar{\omega}^2 x^2 + \frac{1}{2}\frac{c_1^2}{\omega_1^2}x^2 - c_1 q_1 x + \frac{1}{2}\omega_1^2 q_1^2$$

The Hessian is

$$\mathbf{H} = \begin{pmatrix} -\bar{\omega}^2 + \frac{c_1^2}{\omega_1^2} & -c_1 \\ -c_1 & \omega_1^2 \end{pmatrix}$$

and has eigenvalues

$$\begin{aligned} \lambda_0 &= \alpha/2 - \sqrt{\bar{\omega}^2 \omega_1^2 + \alpha^2/4} \\ \lambda_1 &= \alpha/2 + \sqrt{\bar{\omega}^2 \omega_1^2 + \alpha^2/4} \end{aligned} \quad \alpha = \frac{c_1^2}{\omega_1^2} + \omega_1^2 - \bar{\omega}^2$$

It is clear that $\lambda_0 < 0$ and $\lambda_1 > 0$. Therefore the first corresponds to our reaction coordinate and the second to the perpendicular mode. It is also easy to show that the determinant of $\mathbf{H} = -\bar{\omega}^2 \omega_1^2$, and we know that this must be equal to $\lambda_0 \lambda_1$ as the determinant is always equal to the product of the eigenvalues.

The situation in the reactant well is similar but has $+\frac{1}{2}\omega_0^2 x^2$ instead of $-\frac{1}{2}\bar{\omega}x^2$. Thus $Q_r(T) = (\beta\hbar\omega_0)^{-1} \times (\beta\hbar\omega_1)^{-1}$.

If we were to use the dividing surface along the x -coordinate, we would have the one-dimensional TST rate

$$\begin{aligned} k_{\text{TST,1D}} &= Q_r^{-1} \frac{1}{2\pi\beta\hbar} \frac{1}{\beta\hbar\omega_1} e^{-\beta V^\ddagger} \\ &= \frac{\omega_0}{2\pi} e^{-\beta V^\ddagger} \end{aligned}$$

as before.

However, by using the dividing surface perpendicular to the imaginary normal mode, and then can show that

$$k_{\text{hTST}} = Q_r^{-1} \frac{1}{2\pi\beta\hbar} \frac{1}{\beta\hbar\nu_1} e^{-\beta V^\ddagger}$$

where we have defined the normal-mode frequencies by $\lambda_0 = -\nu_0^2$ and $\lambda_1 = \nu_1^2$. We call this hTST as it employs a harmonic approximation to the multidimensional system and uses the optimal dividing surface for this problem. We will assume that this is a good approximation to the full classical rate.

Finally, we compute the transmission coefficient:

$$\kappa \equiv \frac{k_{\text{hTST}}}{k_{\text{TST,1D}}} = \sqrt{\frac{\omega_1^2}{\lambda_1}} = \sqrt{\frac{\omega_1^2}{\lambda_1} \frac{\lambda_0 \lambda_1}{-\omega_1^2 \bar{\omega}^2}} = \sqrt{\frac{-\lambda_0}{\bar{\omega}^2}} = \frac{\nu_0}{\bar{\omega}}$$

which is known as the Grote–Hynes transmission coefficient.

One can easily plot this function and will find that for weak coupling $c_1 \rightarrow 0$ that $\kappa \rightarrow 1$, whereas for strong coupling $c_1 \rightarrow \infty$ that $\kappa \rightarrow \omega_1^2/c_1$. This is in agreement with Kramers theory in the strong coupling regime (i.e. that the rate decreases as the coupling strength increases) and shows that when used properly on the full system TST can actually describe the rate rather well, even though TST applied only to the x -coordinate is wrong because it can't describe the recrossing in this mode.

9.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Microscopic derivation of the Langevin equation

- Zwanzig, Nonequilibrium statistical mechanics (Oxford), chapter 1
- Tuckerman, Statistical Mechanics (Oxford), chapter 15 [includes information on how to implement Langevin dynamics]
- Nitzan, Chemical Dynamics in Condensed Phases (Oxford), section 8.2.5

Kramers theory

- Baron Peters, Reaction Rate Theory and Rare Events (Elsevier), chapters 16 and 17 <https://www.sciencedirect.com/book/9780444563491/reaction-rate-theory-and-rare-events-simulations> (available through ETH library using VPN)

Hopefully you may be able to read them on the internet.